

# Post sintering Operations

- Sintered P/M parts may need additional processing to impart certain characteristics such as improved corrosion resistance, dimensional tolerances or improved strength by heat treatment.
- P/M parts are subjected to a number of secondary operations after sintering.
- Heat treatment, machining, joining and surface coating operations (painting, plating) are called as secondary operations and are carried out in very much the same way as for conventional cast or wrought products.
- The presence of porosity in majority of the P/M parts, however, necessitates the use of precautionary measures during many of these post sintering operations.

# Important secondary operations

- (i) Sizing
- (ii) Coining
- (iii) Repressing and re sintering
- (iv) Impregnation
- (v) Infiltration
- (vi) Heat treatment
- (vii) Steam treatment
- (viii) Machining
- (ix) Joining
- (x) Plating
- (xi) Other finishing operations – shot peening

# SIZING

- Sizing is used to correct size, warping, distortion and other dimensional defects during sintering.
- Sizing can also be used to improve the surface finish and wear properties of the surface.
- Normally, sizing is done by forcing the sintered components into a die of slightly smaller dimensions, which plastically deforms and smoothens the faces in contact with the die.
- Sizing of parts such as piston, bearings and cams are done to reduce tolerances.

- Positive sizing and negative sizing.
- **Positive sizing** occurs when an oversize component with a hole smaller than the core rod is pressed into the die.
- **Negative sizing** occurs when an undersize component is put into the sizing die and its hole is larger than the core rod.
- The punches are used to apply pressure on the part to reduce the length of the part considerably and consequently pressing the part to the die walls and core rods.
- **A thin-walled bushing which is oversize on both outside and inside diameter after sintering can be subjected to positive sizing on the outside and negative sizing on the inside.**

- In sizing, the amount of deformation is small and hence the force required for deformation is less in comparison to the compaction force.
- The subsequent improvements in density as well as mechanical properties are also limited.

# COINING

- The coining operation carried out after sintering achieves the same results as the sizing operation.
- Significant improvement in density, hardness and strength, as well as decrease in elongation of the part are also obtained because of **higher pressures used compared to sizing.**
- Tooling used for sizing and coining are similar in design as that used for compacting powders.
- **Coining is also used for adding surface patterns on to a sintered part.**

- **Hot coining** consists in the rapid coining of a hot sintered compact directly from the sintering furnace.
- The bright, clean surfaces within the porous sintered compact are compressed and deformed, producing a sound, highly dense body.
- The technique is in general more useful for **cermets** than for other materials.
- The parts may be produced from cermets to very close dimensional tolerances and with excellent properties.



# RE PRESSING and RE SINTERING

- The primary aim of repressing is **densification** of the sintered part to give improved physical and mechanical properties along with dimensional control.
- Most P/M components are repressed to reduce dimensional scatter as well as to obtain **closure of surface pores**.
- The **forces involved in repressing are much higher than in sizing operation** and the resulting deformation is also much more.
- However, the **shape change and the deformation encountered is considerably less than in forging or extrusion** as it adjusts only final dimensions without large shape changes.
- For repressing, the compact and tool sizes are the same and material flow is in one direction only.

- In cases where repressing is to be carried out, the sintering temperature is intentionally kept low to keep the compact weak while the repressing pressure exceeds the original compaction pressure.
- With a repressing treatment, pores near the surface are sealed but internal pores remain mostly unaffected.
- After repressing, there is a slight increase in strength with a reduction in ductility and toughness and fatigue strength improves.
- If repressing is carried out at room temperature, the compacts are sintered again.
- The tolerance limit achievable in repressing and sintering are dependent on material type and part size.

- All the above three secondary pressing can be carried out using **hydraulic or mechanical presses**.
- Tools used are similar to those used for powder compaction, but having shorter punch and core rod lengths due to shorter part and work stroke length.
- Use of external lubricants (liquid or dry) is preferred and care should be taken to avoid retention of lubricant by the part which may interfere with further operations like plating.

# IMPREGNATION

- Impregnation of sintered P/M parts is done using either a lubricant or a resin, the purpose being different in each case.
- Structural parts, which serve as bearings, may be impregnated with a lubricant.
- In the case of porous bearings, oil impregnation is done after sizing.
- The viscosity of the oil and the use of additives to inhibit corrosion, to increase film strength and to inhibit oxidation depend on the specific application of the bearing.
- During impregnation, the bearings are placed in a vacuum chamber and evacuated to nearly 1 mm of Hg pressure.
- Oil is allowed to enter into the chamber.
- Air or nitrogen at a pressure of 0.5 MPa is applied for one minute to complete the impregnation.

- Resin impregnation is widely practiced for valves, pumps, pistons and compressor parts requiring pressure tightness.
- P/M parts to be plated can also be treated by resin impregnation to seal the pores and to avoid corrosion by plating chemicals later.
- Gun components are typically resin impregnated to seal the pores.
- The benefits of impregnation, apart from self-lubricating ability, include increase in mechanical properties, pressure-tightness due to sealing off of the porosity, prevention of the entry of corrosive salts into pores during plating and improvement of machinability.

# INFILTRATION

- Pores of a sintered solid are filled with a liquid metal or alloy.
- The solid must have a melting point considerably higher than that of the infiltrating metal or alloy.
- Infiltration is used to improve the strength of ferrous structural parts.
- Infiltration consists of fabricating a porous skeleton of a refractory base material followed by infiltration with a molten metal to produce a sound, strong, non-porous body.
- Only a few metals are suitable for use as infiltrant—for instance copper and copper alloys, silver and nickel.
- An infiltrant must wet the skeleton completely, have low surface tension so as to fill small interstices and must not alloy or react with the base material.

- A continuous network of a ductile infiltrant metal gives the composite much greater resistance to thermal and mechanical stresses.
- The infiltrant can be injected by a variety of techniques such as capillary dip, contact infiltration, gravity feed and pressure or vacuum infiltration.



- sintered part with infiltrant slug on top, infiltration in progress, and infiltrated part.

- In order to infiltrate a porous skeleton of a solid phase with a liquid phase, the total surface free energy of the system after the infiltration should be lower than the total surface free energy before infiltration.
- A **low-wetting angle, therefore, promotes infiltration** (i.e. ready wetting of the solid by the liquid).
- Infiltration may be further aided by establishing a pressure gradient, for example putting the porous skeleton in a vacuum and applying pressure to the liquid, to infiltrate the porous skeleton.
- Wetting may be promoted by additions of a solid, liquid or gas to the system to be infiltrated.

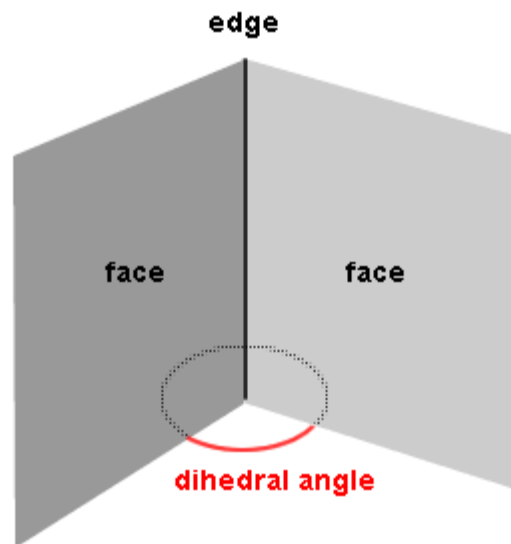


- The infiltration of a porous skeleton with a liquid metal comprises of
  - (i) an actual penetration stage and
  - (ii) a second stage, (after the pore volume has been filled) in which the infiltrant while still liquid, will react with the skeleton.

For a clear understanding of these stages, the following points must be considered:

- The **liquid and solid must not react** to form a solid compound or alloy having a specific volume equal to or greater than their combined pre-infiltration specific volumes, otherwise the reaction product blocks the entry of additional liquid before infiltration is complete.
- The porosity in the **porous skeleton should be interconnected** since the infiltrant cannot penetrate into closed-off pores in the skeleton.

- Ideally, the skeleton material should **be insoluble in the liquid** infiltrant (e.g. copper and silver as infiltrant in tungsten).
- The dihedral angle at the intersection of a boundary of two grains of the solid phase and the liquid phase plays an important role in infiltration.



- If the **dihedral angle is zero**, the liquid will penetrate along grain boundaries in the solid skeleton.
- This may be advantageous because the **liquid may in this way reach closed-off pores in the skeleton**.
- On the other hand, the final properties of the infiltrated material may depend upon the **mechanical properties of the porous skeleton in the as sintered condition**.
- Penetration of the liquid along the grain boundaries of the skeleton may considerably **weaken the original porous skeleton structure**.
- It may also make the control of **dimensional changes during infiltration more difficult**.

# Major advantages of infiltration

- Enhanced mechanical properties—higher tensile strengths, hardness, greater impact energy and fatigue strength.
- Uniform density as infiltration tends to even out variations in density.
- Higher density by increasing the weight of the sintered part without altering its size.
- Removal of porosity which permits secondary operations such as pickling, plating to be carried out.
- Selective-property variation can be achieved by infiltrating only selected areas of a part.
- Assembly of multiple parts by sintering the different parts assembled together and bonding the parts into a single part through common infiltration.

- **Electrical contacts** are among the most important P/M parts produced by infiltration.
- They are most commonly compound materials with a tungsten skeleton infiltrated with copper.
- Combinations of tungsten and silver, molybdenum and silver are other examples of infiltrated materials.
- The tungsten-copper compound materials have also been used as resistance welding electrodes.
- Other infiltrated components include **rocket nozzles** made of tungsten and silver, experimental cermet turbine blades made of titanium carbide-superalloy, tools made of titanium carbide-steels, mechanical parts made of iron/steel-copper alloy and bearings made of steel- babbitt alloy as well as superconducting  $\text{Nb}_3\text{Sn}$  tapes and wires.

# HEAT TREATMENT

- The major heat treatment operations are quenching and tempering, case hardening and annealing in special cases.
- The heat treatment procedure of P/M structural parts, though similar to those of conventional cast or wrought products involves **some basic limitations**.
- While heat-treating P/M steel parts, **salt baths and cyanide baths are generally not used** for austenizing to avoid the possibility of trapping salt in the porous structure.
- By giving carburizing and carbonitriding treatments, the surface hardness of P/M steel parts can be improved.
- In P/M parts, the **carburizing or carbonitriding atmosphere penetrates into the pores and produces a case depth as a function of the part density**.
- Heat-treating can improve strength and hardness and make the surface wear-resistant.

- Many of the P/M structural parts are subjected to quenching and tempering.
- Important considerations in quenching and tempering of P/M parts include the **dependence of heat treatment response on the combined carbon** (rather than on total carbon), the influence of sintering time and temperature on the amount of carbon to be added.
- **Austenizing is usually done with slightly higher temperatures and much longer times (as high as 50% higher) than conventional parts** because of the limited homogeneity of the P/M parts.
- The austenizing time is also dependent on the part section thickness.

- Furnaces used include both batch and continuous type, but the choice depends on the size of the parts, and total part quantity.
- Usually an endothermic and sometimes vacuum atmosphere are used for heating the parts.
- Typical quenching media employed are water (plain water, brine etc.) and oil. However, oil is the preferred medium since it avoids cracking (due to drastic cooling) as well as corrosion.
- Tempering of quenched parts is done immediately in a forced air circulation furnace with temperatures not below 150°C to obtain a tempered martensitic structure.
- Annealing is done mainly for P/M forgings to improve machinability.



- P/M parts are surface hardened by carburizing, carbonitriding, nitrocarburizing and nitriding processes.
- Gas carburizing and pack carburizing processes are used.
- Liquid carburizing is not used because of difficulties in cleaning.
- Carburizing of P/M parts requires that the parts have a minimum density (90% of theoretical density - 7.2 g/cc in case of steels) to have a good case development.
- In case of low-density parts, gas carburizing may lead to gas penetration.
- Case depth depends on temperature and time of carburization.

- **Carbonitriding of P/M** ferrous parts is suitable for both high and low-density parts.
- The processing temperatures and times are lower than for carburizing.
- The process involves the heating of parts in an atmosphere of carbon and ammonia.
- Nitrogen formed by the dissociation of ammonia diffuses into the part along with carbon.
- Carbonitriding is usually carried out at temperatures of 790°C. Continuous and batch type furnaces are used.
- Parts are oil quenched to obtain a predominantly martensitic structure.
- After heat treatment the parts are tempered at a temperature of 200°C.

- In **nitrocarburizing**, an atmosphere consisting of a mixture of ammonia and endothermic gas is used.
- The treatment is carried out on sintered parts at temperature between 500 and 570°C, much lower than carbonitriding.
- The microstructure of the case (after quenching in oil) consists of a thin layer of iron carbonitride and iron nitride (epsilon nitride, the “white layer”), with a diffused layer of iron nitride, alloy nitrides and nitrogen.
- **Gas nitriding of P/M parts is used only to a limited extent.**
- Nitriding gives best results when the parts have a tempered martensitic structure.

# STEAM TREATMENT

- Steam treatment is a process unique to P/M parts, which involves exposing the part at a temperature around 500 to 550°C to high-pressure steam.
- The steam reacts with the outer surface as well as with the inner surfaces of the open pores exposed to the steam.
- This leads to the formation of a layer of magnetite (iron oxide) on all accessible surfaces and a number of desirable properties change results.
- Filling of the pores increases the corrosion resistance and also improves the compressive strength of the part.
- The oxide layer formed increases the surface hardness and particularly the wear resistance of the part.
- Steam treatment is often followed by dipping in oil, to enhance the blue/black appearance and also to increase the corrosion resistance of the component.

- The treatment is **not generally applicable to hardened and tempered parts** because the exposure to high temperature will result in over-tempering of the part.
- Steam treatment is sometimes used in conjunction with tempering (called **steam tempering**) but used on non-heat treated components as well.
- The process consists of **degreasing the parts**, loading them into trays into the furnace, heating the furnace to a temperature of 350°C, introducing the steam at full pressure.
- The temperature is then raised to 370°C and steam is purged for 15 minutes.
- The furnace temperature is then raised to 540°C and held for 90 minutes with steam on.
- The parts are then cooled to 290°C, steam is turned off and the parts are removed.
- Quenching of the parts in oil is done sometimes to improve surface finish and corrosion resistance.

- **Blueing** refers to heating the P/M part in air at a lower temperature (200–250°C) to provide **a thin magnetite layer** that gives some increase in corrosion resistance, though **not as effective as steam treatment**.

# MACHINING

- P/M structural parts are generally designed in such a way that the sintered parts have the specified dimensions.
- Hence P/M parts generally **require no machining**.
- However, in certain cases, machining may be required.
- For example, when holes at an angle to the direction of pressing or tapped holes are needed, machining is done.
- Machining of P/M parts is also carried out to add **holes, threads, undercuts, grooves as well as special features that cannot be incorporated into the die design**.

- The presence of porosity alters the machining characteristics and in **general tool wear is greater** than with the same composition in the fully dense form.
- **Porous P/M parts exhibit a low-thermal conductivity** and hence the cutting tool life is short due to heat erosion.
- In addition, the **stress at the tip of the cutting tool is also cyclic**.
- This is due to the fact that the tool is stressed while in contact with solid and the stress is relieved in contact with a pore.
- This cyclic loading, can lead to **fatigue failure** of the tool machining porous parts, especially when the pores are open and connected.
- This results in shorter tool life compared to a tool that experiences constant cutting force.



- The **machinability can be improved by incorporating certain additions** in the powder mix, e.g. lead, copper, graphite, sulphur or a metal sulphide such as manganese sulphide and also by infiltration or resin impregnation.
- Attempts have been made to improve the tool life and surface finish of P/M steel parts by adding sulphur in the powder compact.
- Addition of lead to brass improves the machining of the parts made of brass.
- Additives such as manganese sulphide, bismuth, tin, tellurium, lead and molybdenum compounds may also be incorporated to improve machinability, as these additives tend to coat the cutting tool and provide lubrication.
- **Material hardness and residual porosity also affect tool life.**
- Longer tool life is achieved when the porosity is either very high or very low.
- In case of very low-density parts, **tool life may be improved by infiltrating the pores with a low- melting metal before machining.**
- **If these points are taken into account, all the traditional machining processes can be done on P/M parts.**

- Carbide tools are recommended for machining operations and lower cutting speeds may be necessary.
- A number of operations such as boring, broaching, burnishing, drilling, grinding, hobbing, honing, lapping, milling, planing, reaming, shaping, tapping, turning and Electro- Discharge Machining (EDM) may be carried out on P/M parts.
- Surface finish obtained after machining P/M parts is never comparable with those of wrought parts.
- The recent availability of multiaxis computer controlled tool control systems have enabled high- precision fabrication of P/M parts free from cracks and machining of components with the most intricate geometry.
- Suitable choice of cemented carbide tool grades and proper choice of the grinding wheels can also improve the machining and grinding characteristics of P/M parts.
- When cutting fluids are used for machining of P/M parts, care must be taken to reduce contamination due to fluids entrapped in the pores by thorough cleaning.

- **Deburring** is another operation done with sintered parts for removing any 'rag' on edges, resulting from the compacting operation or a machining step.
- **Tumbling**, sometimes in a liquid medium with an abrasive powder, is also employed.
- During this operation, brushing and some hardening of the surface layers may also take place.

# JOINING

- Joining of P/M parts is necessitated by the fact that some shapes are not readily made by P/M processing.
- Most of the conventional welding methods for wrought materials are applicable to P/M structural parts, though **porosity is a hindrance**.
- The major metal joining processes applicable for P/M parts include resistance projection welding, gas tungsten arc and gas metal arc welding, electron beam welding, laser beam welding and friction welding.

- Among the various welding operations, **electrical resistance welding is preferred to oxyacetylene and arc welding** to avoid the possibility of oxidation of the interiors of the porous P/M parts in the latter methods.
- Stainless steel P/M parts have been argon arc welded successfully.
- Copper infiltrated parts can be copper brazed by combining brazing and infiltration as a single operation.
- By **combining a composition which expands on sintering (as inner portion) along with another composition, which shrinks during sintering, excellent joints can be formed in the final product.**
- Multiple P/M parts may be joined using diffusion or sinter bonding.
- **Presence of porosity may aggravate the cracking tendency during welding and many of the commercial welding techniques are not readily applicable.**
- Generally, parts with more porosity cannot be welded, though repressed parts may be welded.
- **Major precautions as in the case of welding of wrought parts, such as preheating and post heating in case of high hardenability and limiting the heat affected zone width are a must.**

- **Brazing** of P/M parts is also practiced for some applications.
- **Furnace brazing** is the most common method of brazing P/M parts.
- The presence of porosity in the parts may cause the filler material to be absorbed in the pores.
- This may be avoided by **brazing after infiltrating or after repressing**.
- Brazing with copper is very common.
- Typical atmospheres used include lean exothermic, rich exothermic or even vacuum.

# PLATING

- P/M structural parts can be **electroplated** or by other plating methods.
- Copper, nickel, cadmium, zinc and chromium plating can be done for P/M parts.
- **Porous parts must be resin impregnated** before plating so as to avoid penetration and entrapment of plating solutions in the pores.
- These chemicals, if entrapped, may subsequently ooze out causing blistering of plated layer.
- In case of oil-quenched parts, oil must be removed before resin impregnation.
- Use of **electroless nickel plating can be an alternative solution** for plating of **unimpregnated porous parts**.
- This ensures that the surfaces of the open pores are also coated.

# OTHER COATINGS

- A large percentage of hard metals cutting tool inserts are now coated using Chemical Vapour Deposition (CVD) or Physical Vapour Deposition (PVD).
- The lower temperature PVD process also allows steels to be given a wear-resistant layer of titanium carbide, titanium nitride, alumina or a combination of these materials.
- Sintered high-speed steel tools are also now being coated.